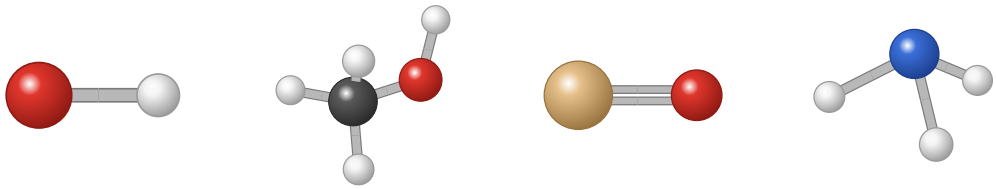
linecat

Viewer for CDMS spectral line catalogs



User Guide

Version 0.6.0

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https://github.com/vahid-anari/linecat

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1 Getting started

linecat ("line catalog") shows the spectral line catalogs of the Cologne Database for Molecular Spectroscopy (CDMS) as a table you can filter by quantum numbers and frequency, and draws the energy levels and transitions of any selection as a diagram that can be saved as PDF or exported as LaTeX commands.

1.1 Installing

linecat needs Python 3.10 or newer. From a clone of the repository:

```
git clone https://github.com/vahid-anari/linecat.git
cd linecat
pip install -e .
linecat
```

LaTeX is optional: with a LaTeX installation the filter labels and the diagram can be rendered with real LaTeX (Section 3.4); without it linecat uses Matplotlib's built-in math rendering.

1.2 First start and the built-in catalogs

linecat ships the CDMS catalog files of four molecules: OH, CH_3OH, SiO and NH_3. On the first start, and after every update of linecat, it converts them into CSV catalogs; this takes about a second. The CSV files are written read-only to the application-data folder (on macOS `~/Library/Application Support/linecat/catalogs/`); if one goes missing it is rebuilt at the next start.

Open one with File ▷ Open Built-in Catalog or with Cmd+1–Cmd+4 (OH, CH_3OH, SiO, NH_3). File ▷ Open Built-in Catalog ▷ Show Built-in Catalogs Folder shows the folder in the Finder.

Note

Any other catalog in the same CSV format opens with File ▷ Open Catalog… (Cmd+O); the last five are listed under File ▷ Open Recent. Section 4 explains how to make such a CSV from a CDMS `.cat` file.

2 The main window

The top row holds three boxes: the molecule, the quantum-number filters and the frequency range. Below them is the table of the lines that pass the filters. The status bar shows how many lines are shown out of how many in the catalog, and the path of the catalog.

2.1 Molecule

The *Molecule* box shows a picture of the molecule, its name, the CDMS tag(s) of the catalog entries, the partition function Q at 300 K used for the line strengths, and the number of lines. Documentation lists the CDMS documentation of the entries (for OH: $v = 0$, 1 and 2); it is included with linecat and opens without an internet connection. The general CDMS documents are under Help ▷ CDMS Documentation.

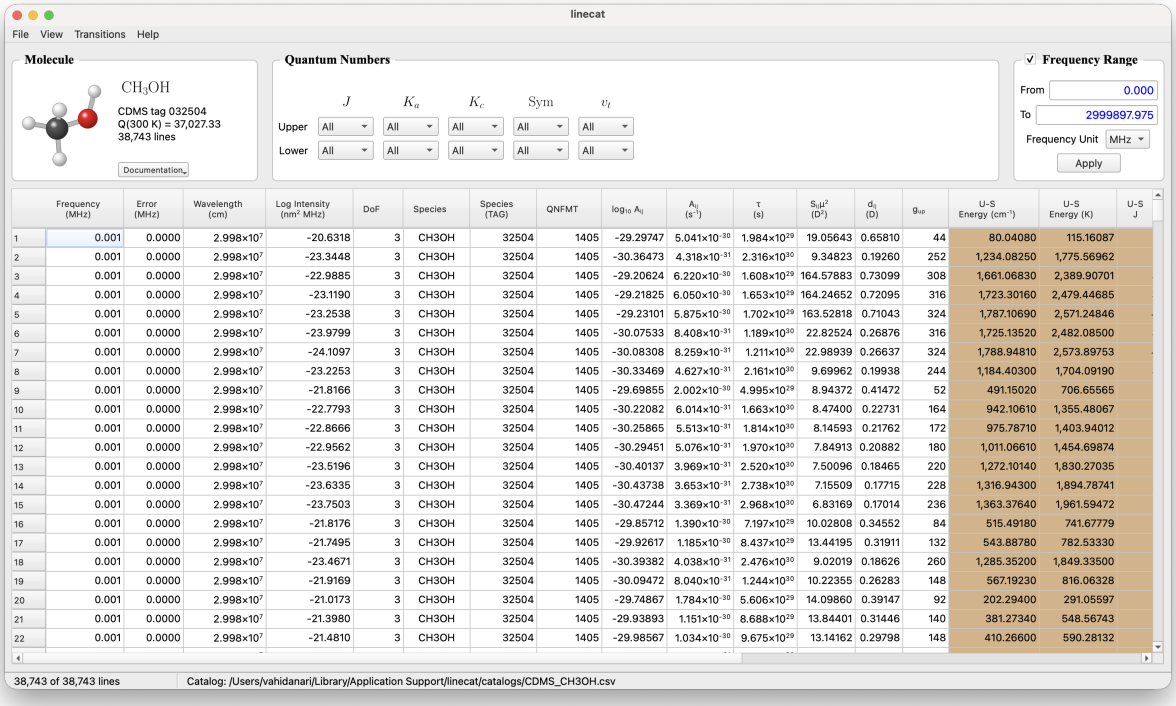


Figure 1: The main window with the built-in CH_3OH catalog.

2.2 Quantum-number filters

Quantum Numbers has one column per quantum number of the molecule and two rows, for the upper and the lower state. Each list offers the values that still occur among the lines that pass the other filters, so the choices narrow as you select. All switches a filter off. View ▷ Reset Filters sets every filter back to All and the frequency range back to the whole catalog.

2.3 Frequency range

From and *To* set the range in the unit chosen under *Frequency Unit*: Hz, KHz, MHz, GHz or THz, or a wavelength in Å, nm, µm, mm or m. Apply applies the range. The check box in the title switches the range off without losing the values typed in.

2.4 The table

Click a column header to sort by that column; click again to reverse the order. Right-click a header for the column menu:

- Column Color…, No Column Color, Default Column Color;
- Text Color…, Default Text Color;
- Alignment (left, centre, right or the default);
- Hide This Column, Choose Columns… (also in View);
- Reset Colors;
- Font…, Default Font.

These settings are remembered between sessions. File ▷ Export Table as CSV… (Cmd+E) writes the table as shown: the filtered lines, the visible columns and the current sort order.

3 The energy-level diagram

Transitions ▷ Show Energy-Level Diagram (Cmd+D) draws the lines shown in the table as a diagram of their energy levels and transitions. Above 50 transitions linecat asks before drawing, since large diagrams become crowded and slow.

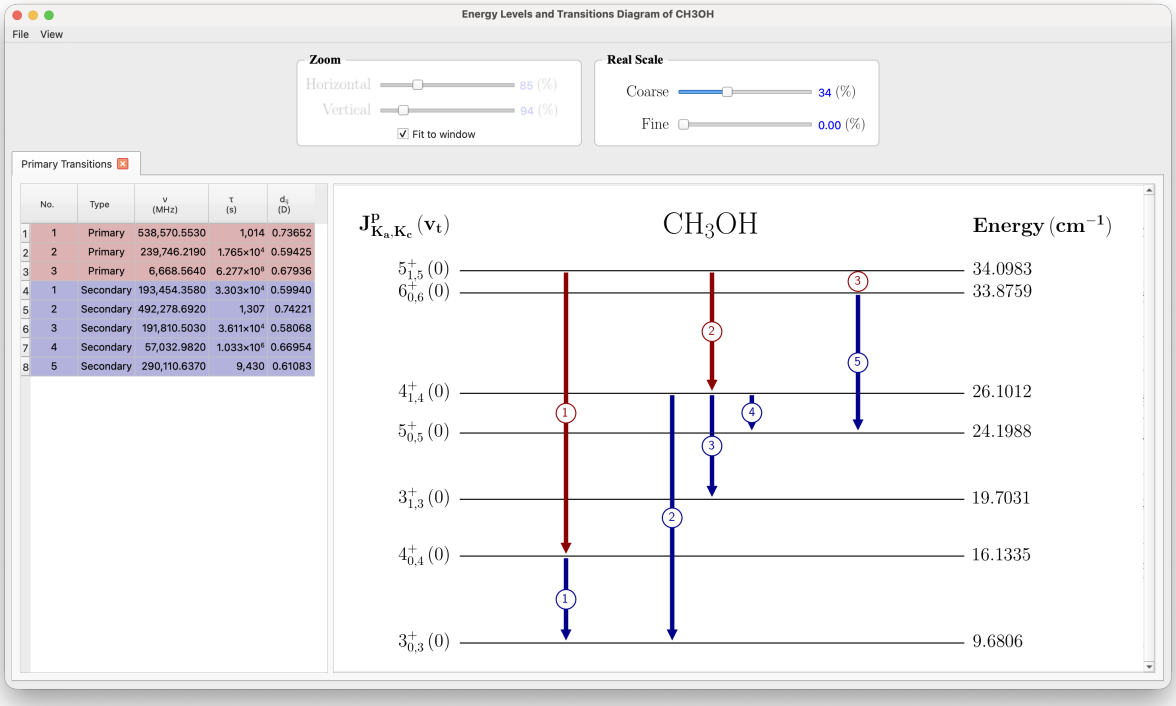


Figure 2: The diagram window: transition table on the left, diagram on the right.

3.1 Levels, transitions and the transition table

Each level is a horizontal line, labelled with its quantum numbers on the left and its energy (cm^{-1}) on the right; each transition is a numbered arrow from the upper to the lower level. The table on the left lists the same transitions with their frequency ν, lifetime τ and transition dipole moment d_{ij}; rows are coloured by type.

Hovering over an arrow highlights it, its two levels and its row in the table, and shows ν and τ next to the pointer; hovering over a level highlights the transitions that start or end on it. The other way round, selecting rows in the table highlights their arrows.

3.2 Tabs and secondary transitions

Double-click an arrow to open a new tab with all transitions that start from its lower level. View ▷ Secondary Transitions adds to the current tab the transitions that leave the lower levels of the

primary ones (drawn in blue). Each tab keeps its own settings; the first tab, *Primary Transitions*, cannot be closed.

3.3 Zoom and real scale

Fit to window (check box under the zoom sliders, or View ▷ Fit to Window, Cmd+0) scales the diagram to the space beside the table and follows the window when it is resized. Switch it off to set the *Horizontal* and *Vertical* zoom by hand.

By default the levels are evenly spaced, in order of energy. The *Real Scale* sliders move them towards their true energies: *Coarse* (0–95 %) and *Fine* (0–5 %) add up to the fraction of the true spacing used, 100 % being energies to scale.

3.4 LaTeX rendering

View ▷ Use LaTeX for Rendering redraws the diagram with LaTeX. The labels are rendered in the background behind a progress dialog the first time; LaTeX keeps the results, so later drawings are quick. In the main window, View ▷ Use LaTeX for Labels does the same for the filter labels.

3.5 Saving

File ▷ Save Figure as PDF… (Cmd+S) writes a two-page PDF: the diagram, then the transition table. File ▷ LaTeX Commands… shows the diagram as LaTeX commands (`\DrawLevel` and `\DrawTr`) to paste into your own document; Copy puts them on the clipboard.

4 Converting CDMS catalogs

The built-in catalogs cover OH, CH_3OH, SiO and NH_3. To use a newer CDMS release, download the `.cat` files of the entries from https://cdms.astro.uni-koeln.de/ and put them in a folder per species, named as in linecat (`OH/c017507.cat`, …). Then choose File ▷ Convert CDMS Catalogs… and select either the folder that holds the species folders or a single species folder. The CSV files are written next to the `.cat` files; a progress dialog and a short report follow. A single converted catalog opens by itself.

For each line the converter computes the wavelength, the upper-state energy, both energies in kelvin, the Einstein coefficient A_{ij}, the lifetime $\tau = 1/A_{ij}$, the line strength $S_{ij}\mu^2$ and the transition dipole moment d_{ij}, using the relations in Appendix A.

5 Command line

`linecat`	open the viewer
`linecat CATALOG.csv`	open a catalog at start
`linecat convert`	convert the built-in CDMS files (into the current folder)
`linecat convert OH SiO`	convert some species
`linecat convert -catalogs DIR`	convert your own `.cat` files in DIR
`...  -output-dir OUT`	write the CSV files to OUT
`linecat -version`	show the version

6 Adding a species

A species is described in `linecat/molecules.py` by a `MoleculeInfo`: its name, the `.cat` files, the partition function and temperature, the quantum numbers of the CDMS format and the filters shown in the window. Two functions complete it: `<name>_standardize(df, prefix)`, which turns the raw quantum numbers into the ones shown (for example Ω and the parity of OH), working on whole columns of the table, and `<name>_label(df, prefix)`, which writes the level labels of the diagram. Register the species in `REGISTRY` and put its `.cat` files, the CDMS documentation PDFs of its entries and a picture `<SPECIES>.svg` in `linecat/data/molecules/<SPECIES>/`.

References

The catalog data come from the Cologne Database for Molecular Spectroscopy; please cite

- H. S. P. Müller et al., 2001, Astron. Astrophys. 370, L49;
- H. S. P. Müller et al., 2005, J. Mol. Struct. 742, 215;
- C. P. Endres et al., 2016, J. Mol. Spectrosc. 327, 95.

The catalog format and the SPCAT program are described by H. M. Pickett et al., 1998, J. Quant. Spectrosc. Radiat. Transfer 60, 883. The CDMS documentation pages included with linecat are reproduced with credit to CDMS.

A Equations

These are the relations `linecat convert` uses to derive the radiative quantities of each line from the CDMS catalog entries (frequency, intensity at $T = 300$ K, lower-state energy, upper-state degeneracy and the partition function $Q_{rs}(300\,\mathrm{K})$). All equations are in cgs units.

A.1 Constants and conversion factors

$$\begin{aligned} h &= 6.62607015 \times 10^{-27}\,\mathrm{erg\cdot s} \\ k_B &= 1.380649 \times 10^{-16}\,\mathrm{erg\cdot K^{-1}} \\ c &= 2.99792458 \times 10^{10}\,\mathrm{cm\cdot s^{-1}} \\ 1\,\mathrm{D} &= 10^{-18}\,\mathrm{statC\cdot cm} \\ 1\,\mathrm{erg} &= \mathrm{statC^2\cdot cm^{-1}} \\ {}^{x}S_{ij}\,\mu_x^2 &= g_i\,\mu_{ij}^2 \end{aligned} \tag{1}$$

A.2 Conversion relations (Section A.6)

Frequency ($\mathrm{MHz} \to \mathrm{cm^{-1}}$):

$$\left[\frac{\tilde{\nu}}{\mathrm{cm^{-1}}}\right] = 3.33564095 \times 10^{-5} \left[\frac{\nu}{\mathrm{MHz}}\right] \tag{2}$$

Energy ($\mathrm{cm^{-1}} \to \mathrm{K}$):

$$\left[\frac{T}{\mathrm{K}}\right] = 1.43877709 \left[\frac{\tilde{\nu}}{\mathrm{cm^{-1}}}\right] \tag{3}$$

A.3 Intensity

The intensity $I(T)$ is calculated according to

$$I(T) = \frac{8\pi^3}{3hc}\,\nu\,S_g\,\mu_g^2\,\frac{e^{-E''/kT} - e^{-E'/kT}}{Q_{rs}(T)} \tag{4}$$

with ν and S_g the line frequency and strength, μ_g the dipole moment along the molecular g-axis, E'' and E' the lower and upper state energy, and Q_{rs} the rotation-spin partition function at the temperature T. With $I(T)$ in $\mathrm{nm^2\,MHz}$, ν in MHz and μ_g in D one obtains (Section A.7)

$$I(T) = \nu\,S_g\,\mu_g^2\,\frac{e^{-E''/kT} - e^{-E'/kT}}{Q_{rs}(T)}\;4.16237896 \times 10^{-5}\,\mathrm{nm^2\,MHz} \tag{5}$$

A.4 Spontaneous emission rate

The average spontaneous emission rate from the upper states into the lower states is

$$A = \frac{8\pi\nu^2}{c^2}\,I(T)\left(\frac{Q_{rs}(T)}{g_{up}}\right)\left[e^{-E''/kT} - e^{-E'/kT}\right]^{-1} \tag{6}$$

in which g_{up} is the degeneracy of the upper state. With $I(T)$ in $\mathrm{nm^2\,MHz}$ and ν in MHz one obtains (Section A.7)

$$A = I(T)\,\nu^2\,[Q_{rs}(T)/g_{up}]\left[e^{-E''/kT} - e^{-E'/kT}\right]^{-1}\;2.79639459 \times 10^{-16}\,\mathrm{s^{-1}} \tag{7}$$

Note: numerical problems may occur in this equation if the frequency ν is small with respect to the lower state energy E''. So, if $e^{-E''/kT} - e^{-E'/kT} = 0$ numerically, we use

$$e^{-E''/kT} - e^{-E'/kT} \simeq \frac{h\nu}{kT}\, e^{-E''/kT} \tag{8}$$

A.5 Transition dipole moment

$$\mu^2 = \frac{3hc^3}{64\pi^4\nu^3}\, A \tag{9}$$

With A in s^{-1} and ν in MHz one obtains (Section A.7)

$$\mu = \sqrt{\frac{A}{\nu^3}}\, 9.26894 \times 10^9\,\mathrm{D} \tag{10}$$

A.6 Derivation of the conversion relations

$$\left[\frac{\tilde{\nu}}{\mathrm{cm}^{-1}}\right] = \frac{1}{c}\left[\frac{\nu}{\mathrm{Hz}}\right] = 3.33564095 \times 10^{-5}\left[\frac{\nu}{\mathrm{MHz}}\right]$$

$$\left[\frac{\tilde{\nu}}{\mathrm{cm}^{-1}}\right] = \frac{1}{hc}\left[\frac{E}{\mathrm{erg}}\right] = 5.03411657 \times 10^{15}\left[\frac{E}{\mathrm{erg}}\right]$$

$$\left[\frac{T}{\mathrm{K}}\right] = \frac{1}{k_B}\left[\frac{E}{\mathrm{erg}}\right] = \frac{hc}{k_B}\left[\frac{\tilde{\nu}}{\mathrm{cm}^{-1}}\right] = 1.43877709\left[\frac{\tilde{\nu}}{\mathrm{cm}^{-1}}\right]$$

A.7 Derivation of the radiative quantities

Here i is the upper and j the lower state.

Intensity

$$\begin{aligned}
I_{ij}(T) &= \frac{8\pi^3}{3hc}\,\nu_{ij}\,{}^{x}S_{ij}\mu_x^2\,\frac{e^{-E_j/kT} - e^{-E_i/kT}}{Q_{rs}(T)} \\
&= 4.16237896 \times 10^{17}\left(\frac{\nu_{ij}}{\mathrm{Hz}}\right)\left(\frac{{}^{x}S_{ij}\mu_x^2}{\mathrm{statC}^2\cdot\mathrm{cm}^2}\right)\frac{e^{-E_j/kT} - e^{-E_i/kT}}{Q_{rs}(T)}\,\frac{\mathrm{Hz}\cdot\mathrm{statC}^2\cdot\mathrm{cm}^2}{\mathrm{erg}\cdot\mathrm{s}\cdot\mathrm{cm}\cdot\mathrm{s}^{-1}} \\
&= 4.16237896 \times 10^{17}\left(\frac{\nu_{ij}}{\mathrm{Hz}}\right)\left(\frac{{}^{x}S_{ij}\mu_x^2}{\mathrm{statC}^2\cdot\mathrm{cm}^2}\right)\frac{e^{-E_j/kT} - e^{-E_i/kT}}{Q_{rs}(T)}\,\mathrm{cm}^2\,\mathrm{Hz} \\
&= 4.16237896 \times 10^{31}\left(\frac{\nu_{ij}}{\mathrm{MHz}}\right)\left(\frac{{}^{x}S_{ij}\mu_x^2}{\mathrm{statC}^2\cdot\mathrm{cm}^2}\right)\frac{e^{-E_j/kT} - e^{-E_i/kT}}{Q_{rs}(T)}\,\mathrm{nm}^2\,\mathrm{MHz} \\
&= 4.16237896 \times 10^{31}\left(\frac{\nu_{ij}}{\mathrm{MHz}}\right)\left(\frac{{}^{x}S_{ij}\mu_x^2}{\mathrm{D}^2}\,\frac{\mathrm{D}^2}{\mathrm{statC}^2\cdot\mathrm{cm}^2}\right)\frac{e^{-E_j/kT} - e^{-E_i/kT}}{Q_{rs}(T)}\,\mathrm{nm}^2\,\mathrm{MHz} \\
&= 4.16237896 \times 10^{-5}\left(\frac{\nu_{ij}}{\mathrm{MHz}}\right)\left(\frac{{}^{x}S_{ij}\mu_x^2}{\mathrm{D}^2}\right)\frac{e^{-E_j/kT} - e^{-E_i/kT}}{Q_{rs}(T)}\,\mathrm{nm}^2\,\mathrm{MHz}
\end{aligned}$$

$$\left[\frac{I_{ij}(T)}{\mathrm{nm}^2\,\mathrm{MHz}}\right] = 4.16237896 \times 10^{-5}\left(\frac{\nu_{ij}}{\mathrm{MHz}}\right)\left(\frac{{}^{x}S_{ij}\mu_x^2}{\mathrm{D}^2}\right)\frac{e^{-E_j/kT} - e^{-E_i/kT}}{Q_{rs}(T)}$$

Einstein A coefficient

$$
\begin{aligned}
A_{ij} &= \frac{64\pi^4\nu_{ij}^3}{3hc^3}\,\mu_{ij}^2 = \frac{64\pi^4\nu_{ij}^3}{3hc^3}\,\frac{{}^x S_{ij}\mu_x^2}{g_i} \\
&= \frac{64\pi^4\nu_{ij}^3}{3hc^3}\,\frac{1}{g_i}\left\{\frac{I_{ij}(T)\,Q_{rs}(T)}{\frac{8\pi^3}{3hc}\,\nu_{ij}\left[e^{-E_j/kT}-e^{-E_i/kT}\right]}\right\} = \frac{8\pi\nu_{ij}^2}{c^2}\,\frac{I_{ij}(T)\,Q_{rs}(T)}{g_i}\left[e^{-E_j/kT}-e^{-E_i/kT}\right]^{-1} \\
&= 2.79639459\times10^{-20}\left[\frac{I_{ij}(T)}{\mathrm{nm^2\,MHz}}\right]\left[\frac{\nu_{ij}}{\mathrm{Hz}}\right]^2\left[Q_{rs}(T)/g_i\right]\left[e^{-E_j/kT}-e^{-E_i/kT}\right]^{-1}\frac{\mathrm{nm^2\,MHz\cdot Hz^2}}{\mathrm{cm^2\,s^{-2}}} \\
&= \left[\frac{I_{ij}(T)}{\mathrm{nm^2\,MHz}}\right]\left[\frac{\nu_{ij}}{\mathrm{MHz}}\right]^2\left[Q_{rs}(T)/g_i\right]\left[e^{-E_j/kT}-e^{-E_i/kT}\right]^{-1}2.79639459\times10^{-16}\,\mathrm{s^{-1}}
\end{aligned}
$$

$$
\left[\frac{A_{ij}}{\mathrm{s^{-1}}}\right] = 2.79639459\times10^{-16}\left[\frac{I_{ij}(T)}{\mathrm{nm^2\,MHz}}\right]\left[\frac{\nu_{ij}}{\mathrm{MHz}}\right]^2\left[Q_{rs}(T)/g_i\right]e^{E_j/kT}\left[1-e^{-h\nu_{ij}/kT}\right]^{-1}
$$

Note: if $e^{-E''/kT}-e^{-E'/kT}=0$ numerically, we use

$$
e^{-E''/kT}-e^{-E'/kT} = e^{-E''/kT}\left(1-e^{-(E'-E'')/kT}\right) = e^{-E''/kT}\left(1-e^{-h\nu/kT}\right) \simeq \frac{h\nu}{kT}\,e^{-E''/kT}
$$

Transition dipole moment

$$
\begin{aligned}
\mu_{ij}^2 &= \frac{3hc^3}{64\pi^4\nu_{ij}^3}\,A_{ij} \\
&= 85.9132065\left[\frac{A_{ij}}{\mathrm{s^{-1}}}\right]\left[\frac{\nu_{ij}}{\mathrm{Hz}}\right]^{-3}\frac{\mathrm{erg\cdot s\cdot cm^3\cdot s^{-3}}}{\mathrm{s\cdot Hz^3}} = 85.9132065\left[\frac{A_{ij}}{\mathrm{s^{-1}}}\right]\left[\frac{\nu_{ij}}{\mathrm{Hz}}\right]^{-3}\mathrm{statC^2\cdot cm^2} \\
&= 8.59132065\times10^{19}\left[\frac{A_{ij}}{\mathrm{s^{-1}}}\right]\left[\frac{\nu_{ij}}{\mathrm{MHz}}\right]^{-3}\mathrm{D^2}
\end{aligned}
$$

$$
\left[\frac{\mu_{ij}}{\mathrm{D}}\right] = \sqrt{8.59132065\times10^{19}\left[\frac{A_{ij}}{\mathrm{s^{-1}}}\right]\left[\frac{\nu_{ij}}{\mathrm{MHz}}\right]^{-3}}
$$